

Graduate Student Research Position Available at University of Pittsburgh: “High Frequency Transformer Design, Modeling, Prototyping, and Testing”

Position Opening: Available Immediately

Position: High Frequency Transformer Design, Modeling, Prototyping, and Testing (PhD or MS)

Position Contact: Prof. Paul Ohodnicki, pro8@pitt.edu, Prof. Brandon Grainger, bmg10@pitt.edu

Anticipated Position Start Date: Fall 2026 or Spring 2027 (Target no later than January 2027)

Position Description:

The Advanced Magnetism for Power and Energy Development Consortium at the University of Pittsburgh invites applications for a graduate student researcher with an emphasis on high frequency transformer design, modeling, and testing. Near-term anticipated project work will include modeling and design of medium and high frequency transformers including electromagnetic performance, isolation, thermals, and other design requirements. Additionally, the project will involve prototyping and testing in collaboration with project partners. The position is funded as a result of a recent [ARPA-E award](#) as well as the [Advanced Magnetism for Power and Energy Development](#) consortium based at the University of Pittsburgh. The graduate student researcher will have an opportunity to collaborate with an interdisciplinary group of undergraduate, graduate, and PhD level researchers in magnetic materials and electric power conversion applications. A senior post-doctoral researcher will also be collaborating closely with the graduate student researcher to provide additional mentoring and training. The graduate student researcher can expect close collaboration with other researchers from industry and national labs including in collaboration with partners at North Carolina State University, National Laboratory of the Rockies, and Hitachi Energy, Department of Defense, and others.

The advisors will be Prof. Paul Ohodnicki and Prof. Brandon Grainger, and the research associate will have full access to facilities available in the Ohodnicki Lab (<https://www.engineering.pitt.edu/OhodnickiLab/>) at the University of Pittsburgh, the extensive shared facilities of the Advanced Magnetism for Power and Energy Development (AMPED, <https://www.engineering.pitt.edu/AMPED/>) and the GRID Institute ([Home | Energy GRID Institute | University of Pittsburgh](#)), amongst other facilities across campus.

More information about the Ohodnicki Lab research focus and interests can be found here:

<https://www.youtube.com/watch?v=7Vn7XJmHGr4&feature=youtu.be>

Successful applicants should display strong interest in the concepts of importance for transformer modeling and design, including concepts and tools such as magnetic equivalent circuits, finite element modeling, applied electromagnetics, and multi-objective optimization methods. Applicants should have an undergraduate degree in electrical engineering, applied physics, or a related field. Prior experience with these concepts and tools is desired but not required as the graduate student will be provided with ample resources and training to develop and apply new skills throughout their degree program in this area.

Application Process:

Interested candidates should contact Prof. Paul Ohodnicki (pro8@pitt.edu) and Prof. Brandon Grainger (bmg10@pitt.edu) with an updated resume at the soonest possible opportunity and apply for the graduate degree program in Electrical and Computer Engineering or Mechanical Engineering and Materials Science at the Swanson School of Engineering using the following link: [Graduate Applications](#)